

The Problem

A digital banking onboarding flow applied identical identity-verification rigor to every applicant, regardless of actual risk. This created two simultaneous costs that traced back to two different stakeholders pulling in opposite directions: the Product team (represented by a fictional PM, Aravind) needed a fast, low-friction signup to protect conversion, while Compliance (represented by a fictional Compliance Lead, Priya) needed thorough scrutiny to manage AML/KYC exposure — including ensuring Politically Exposed Persons (PEPs) received Enhanced Due Diligence rather than the same treatment as every other applicant.

Applying uniform friction to satisfy Compliance frustrated low-risk users into abandoning signup; applying uniform speed to satisfy Product left high-risk applicants under-scrutinized. Neither stakeholder's goal could be fully met without addressing the other's — the two were structurally in tension as long as the system treated every applicant the same way.

The Resolution Approach

The resolution was not a compromise between the two goals, but a redesign that let both be true simultaneously: a Risk-Based Approach (RBA) that tiers applicants into Low, Medium, and High risk using a transparent, deterministic weighted formula (country risk, transaction velocity, verification match rate), with PEP/sanctions matches automatically overriding to High-tier Enhanced Due Diligence regardless of computed score. Low-risk applicants — the majority — could move through an unattended fast path; friction concentrated only where risk was real.

This logic was specified across a full requirements chain: a Business Requirements Document, As-Is/To-Be process maps, a 14-requirement Functional Requirements Document (including an explicit precedence rule, FR-014, governing what happens when a PEP match and a failed verification check apply to the same applicant simultaneously — input quality is resolved before the EDD override is confirmed), a Requirement Traceability Matrix, and 13 user stories with acceptance criteria covering realistic edge cases (expired documents, poor camera quality, sanctions matches, borderline scores).

The Findings

A synthetic dataset of 2,000 applicants was generated to test this design against realistic, intentionally noisy data — including a deliberately constructed population of borderline risk profiles, so the threshold-tuning exercise would have a genuine decision to make rather than a trivial one. (This borderline-profile rate was a synthetic-data design choice to ensure analyzable variance; it does not represent any real-world fraud or risk prevalence statistic.)

Drop-off. The original business case assumed a placeholder drop-off rate (~28%, targeting ~18%) for sizing purposes, explicitly flagged as an assumption rather than a measurement. The actual mock-data analysis found a **21.85% overall drop-off rate**, concentrated most heavily at the document-upload stage (8.25% of all applicants) — directionally consistent with the original

hypothesis that document verification was the heaviest friction point, though the specific percentage differs from the initial placeholder and is reported here as the corrected, real figure.

Time-to-Onboard. This is the project's central result. Risk-tiering produced a clear, real separation in onboarding speed: Low-risk applicants completed onboarding in an average of **4.04 minutes**, Medium-risk in **17.79 minutes**, and High-risk in **57.24 minutes** — roughly a 14x difference between the fastest and slowest tiers. This is direct evidence that the tiering approach successfully concentrates friction on the minority of applicants who warrant it, rather than distributing it uniformly.

Threshold selection. A precision/recall sweep was run across a wide range of candidate thresholds (20-75), not just the FRD's original 70-vs-75 candidates, since those both proved too far into the score distribution's tail to produce a meaningful comparison (under 1.6% of applicants scored above 70). The sweep showed the best statistical balance (F1 score) at threshold 40, but that threshold would route 32.2% of all applicants to High-tier/EDD — operationally unworkable, since it would mean a third of users facing the heaviest friction tier, undermining the project's core goal. **Threshold 50 was selected instead**, routing 15.2% of applicants to High-tier and catching 32.4% of applicants later flagged as genuinely risky, at a precision of 0.436.

Known limitation, stated directly. Even at the operationally chosen threshold, the model misses roughly two-thirds of applicants who are later flagged as risky. This reflects a real limit of the three-factor weighted formula's ability to separate risk from legitimacy on this dataset, not a threshold-tuning failure — no threshold choice fully closes this gap. A future iteration would need to test additional or reweighted signals to improve separation, rather than further threshold adjustment alone.

Data sufficiency. When verification pass rates were broken out by risk tier and country, every High-tier country segment (all 12 countries) had fewer than 15 applicants — too few to support a reliable country-level claim. These rows are visibly flagged in the dashboard rather than presented as findings, since a single applicant can swing a small sample's percentage by several points.

The Outcome

The risk-tiering design successfully demonstrates that Product's and Compliance's goals are not inherently opposed — they were only in conflict under a uniform process. With tiering, the majority of applicants (85.4% of those who completed onboarding) moved through in minutes, while scrutiny concentrated on a smaller, genuinely higher-risk population. The project also surfaced a real, documented limitation in the scoring model's risk-separation power, and a real data-sufficiency gap at the country level for the High tier — both reported as findings rather than smoothed over, since an honest account of what the data does and doesn't support is itself a core part of the deliverable.

